

FANUC Robot **series**

R-30*i*B/R-30*i*B Plus CONTROLLER

CD STATIONARY TRACKING AND RPM CLOCK USER'S GUIDE

MARUBCSTK03151E REV C

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FANUC conducts courses on its systems and products on a regularly scheduled basis at the company's world headquarters in Rochester Hills, Michigan. For additional information contact

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FANUC America Corporation Patent List

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Conventions

WARNING

Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.

CAUTION

Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints.

Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.

- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
- Using a low motion speed, test run the program continuously for at least one full cycle.
- Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personnel also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today's highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high-voltage electrostatic tools.



CAUTION

Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.

- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E–STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released or pressed harder during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E–STOP other robots.

 WARNING

An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.
- When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

 WARNING

Observe all safety rules and guidelines to avoid injury.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

 WARNING

Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom-shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.
 1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
 2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
 3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
 4. When the dust has been removed, the covers must be replaced immediately.
 5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.



WARNING

When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

- **Grounding:** All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage:** High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.

- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the “home” position before you disarm it.
- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot’s movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
 - Wear eye protection and protective gloves during application and removal.
 - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
 - If there is contact with the skin, wash with water.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.
- When using paint remover observe the following:
 - Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
 - Adequate ventilation is required. Overexposure could cause drowsiness.
 - If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then seek medical attention as soon as possible.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.

TABLE OF CONTENTS

1.	Function Overview	1
2.	Function Specifications	1
3.	Function Limitation	1
4.	Function Installation	1
5.	Work Cell Configuration Specification (Orthogonal)	2
5.1	UTOOL Relative (\$avst_trcst[sch_no].\$i2 = 0)	2
5.2	UFRAME Relative (\$avst_trcst[sch_no].\$i2 = 1)	2
5.3	Auto Setup \$cell_cfg	5
6.	Tracking Bias Specification	6
6.1	Steps to determine \$bias_z for UFRAME relative cell	6
6.1.1	Example 1 (cell_cfg = 2)	6
6.1.2	Example 2 (cell_cfg = 3)	6
6.1.3	Example 3 (cell_cfg = 6)	7
7.	Stationary Tracking Calibration	7
7.1	Ten Point Sensor Calibration.	7
7.2	Sensor Leading Angle Calibration	7
7.3	Coordinated Motion Setup	9
7.3.1	1-Axis Positioner	9
7.3.2	2-Axes Positioner	9
7.4	Sensor Scan and Registration Rate Control	10
8.	Tack Avoidance and Control	10
8.1	Tack Recognition	10
8.2	Tack Length Measurement	11
8.3	Tack Alarm	11
8.4	Real Time Tack Control	11
8.5	Overlap Welding Control	12
9.	Coordinated Stationary Tracking Setup	13
10.	RPM/MP Clock Function Specifications and Restrictions	15
11.	RPM/MP Clock Operation	15
11.1	Example 1 torch at 12 clock position	15
11.2	Example 2 torch at 1 clock position	17
12.	RPM/MP Clock Setup	18
13.	Coordinated Stationary Tracking and RPM/MP Clock Programming	19
13.1	PR_CLK_G13 Utility TP	20
13.2	Root Pass Program Example	21
13.3	Multiple Pass Program Example	22

14.	Alarms	23
14.1	Alarms for CD Stationary Tracking	23
14.2	Alarms for RPM/MP Clocking	23
15.	Setup Existing Sensor Schedule & Search Schedule	24

Figure 1 \$cell_cfg = 1	2
Figure 2 \$cell_cfg = 2	2
Figure 3 \$cell_cfg = 3	3
Figure 4 \$cell_cfg = 4	3
Figure 5 \$cell_cfg = 5	3
Figure 6 \$cell_cfg = 6	3
Figure 7 \$cell_cfg = 7 (invert mount).....	4
Figure 8 \$cell_cfg = 8 (invert mount).....	4
Figure 9 \$cell_cfg = 9 (invert mount).....	4
Figure 10 \$cell_cfg = 10 (invert mount).....	4
Figure 11 \$cell_cfg = 11 (invert mount).....	4
Figure 12 \$cell_cfg = 12 (invert mount).....	5
Figure 13 Stationary Position at 12 o'clock.....	7
Figure 14 Stationary Position at 11 o'clock.....	8
Figure 15 Stationary Position at 1 o'clock.....	8
Figure 16 SETUP Coord Sreen (single axis positioner)	9
Figure 17 2-Axes Positioner SETUP for axis 1	9
Figure 18 2-Axes Positioner SETUP for axis 2	10
Figure 19 TACK is detected	11
Figure 20 TCP at TACK Section	12
Figure 21 Scans at welded joint	12
Figure 22 Root Pass, Torch at 12 clock position	15
Figure 23 MPass 1, Torch at 12 clock position	16
Figure 24 MPass 2, Torch at 12 clock position	16
Figure 25 Root Pass, Torch at 1 clock position	17
Figure 26 MPass 1, Torch at 1 clock position	17
Figure 27 MPass 2, Torch at 1 clock position	18

1. Function Overview

The Coordinated Stationary Tracking and RPM/MP Clock Function are used with laser sensor seam trackers to provide an automation solution for robotic pipe welding application. It consists of Stationary Tracking function for tracking pass and RPM/MP Clock function for multiple passes. In a tracking pass, robot stays at a stationary position while the pipe is rotating around a rotation axis in coordinated motion. The robot torch is dynamically adjusted in the lateral and vertical directions based on laser sensor feedback information. The applied position offset is stored in a RPM buffer relative to pipe rotation angles. In multiple welding passes, RPM/MP Clock function allows each pass to start at different pipe rotation angle while the offset stored in the rpm buffer can be applied correctly. The RPM/MP clock function is useful for staggered welds for pipe applications and improves weld quality.

2. Function Specifications

- Support RPM/MP with distance and time based pitch type
- Support weave operation
- Support existing 10 point Sensor Calibration Method
- Support 10 tracking schedules
- Support 10 rpm buffers
- Support RPM/MP Clocking function w/ overlap weld
- Support multiple coordinated pairs
- Support dynamic switching between coordinated pairs w/o re-powering controller
- Support dynamic switching between Stationary Tracking and Non-Stationary tracking w/o re-powering controller
- Support stationary tack avoidance and tack control

3. Function Limitation

- Constant Path is disabled.
- Requires coordinated motion option.

4. Function Installation

This software function is bundled to existing laser tracking function and will be loaded automatically with laser tracking option. This function is available beginning in v8.30P06 release.

5. Work Cell Configuration Specification (Orthogonal)

Stationary tracking supports UTOOL and UFRME relative cell configurations.

5.1 UTOOL Relative (\$avst_trcst[sch_no].\$i2 = 0)

UTOOL relative requires pipe rotation axis to be parallel to one of the major robot axes in $\pm X$, $\pm Y$, or $\pm Z$. The cell configuration is determined by Path-Y in lateral direction relative to robot base. The Path-Z follows TOOL-Z direction and Path-X, Path-Y and Path-Z coordinates frame follows right-hand rule. It supports configurations 1-6 but it does not support invert mount configurations 7-12. The UTOOL relative cell configuration is useful when stationary welding position is required to be at any pipe clock positions.

5.2 UFRAME Relative (\$avst_trcst[sch_no].\$i2 = 1)

UFRME relative requires tracking frame to be parallel to robot base. It can support more configurations such as invert or wall mount robot. It supports configurations 1- 12.

Figure 1 \$cell_cfg = 1

Path-Y (lateral) is parallel to (-Y) of robot base

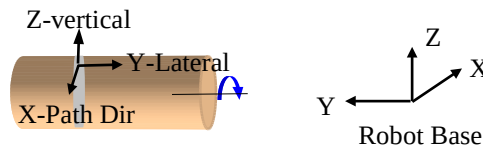


Figure 2 \$cell_cfg = 2

Path-Y (lateral) is parallel to (Y) of robot base

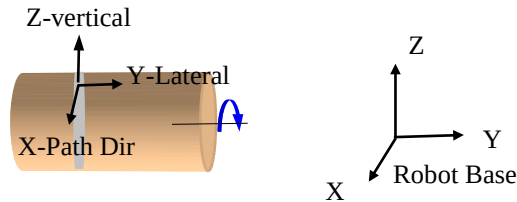
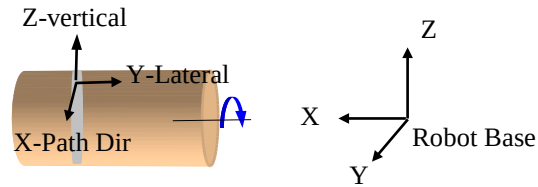
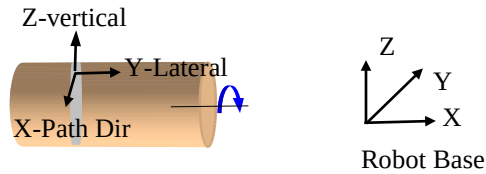


Figure 3 \$cell_cfg = 3

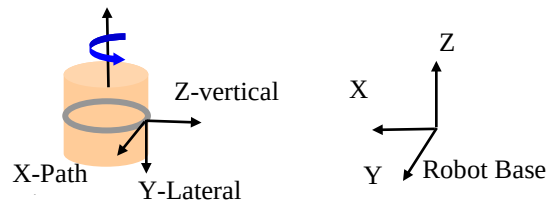
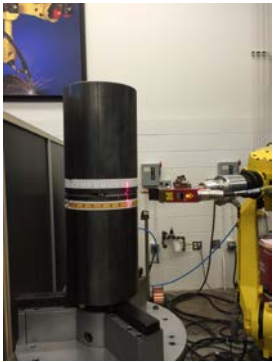
Path-Y (lateral) is parallel to $(-X)$ of robot base

**Figure 4 \$cell_cfg = 4**

Path-Y (lateral) is parallel to (X) of robot base

**Figure 5 \$cell_cfg = 5**

Path-Y (lateral)) is parallel to $(-Z)$ of robot base

**Figure 6 \$cell_cfg = 6**

Path-Y (lateral) is parallel to (Z) of robot base

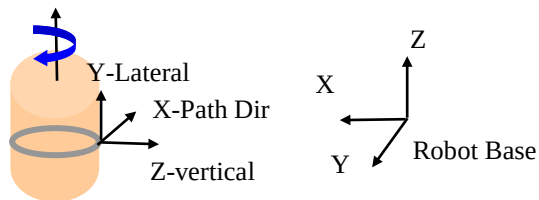
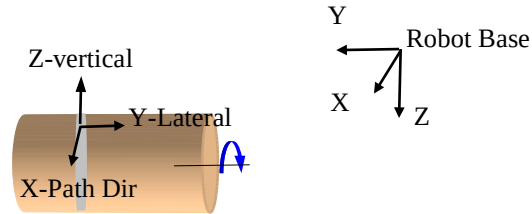
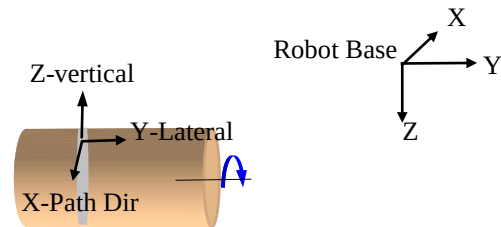


Figure 7 \$cell_cfg = 7 (invert mount)

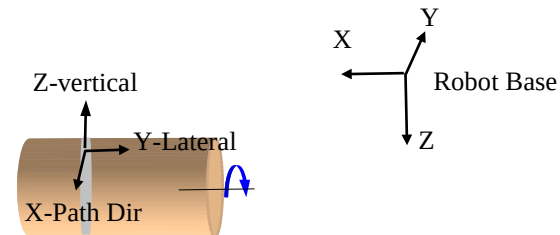
Path-Y (lateral) is parallel to (-Y) of robot base
 Path-Z (vertical) is parallel to (-Z) of robot base

**Figure 8 \$cell_cfg = 8 (invert mount)**

Path-Y (lateral) is parallel to (Y) of robot base
 Path-Z (vertical) is parallel to (-Z) of robot base

**Figure 9 \$cell_cfg = 9 (invert mount)**

Path-Y (lateral) is parallel to (-X) of robot base
 Path-Z (vertical) is parallel to (-Z) of robot base

**Figure 10 \$cell_cfg = 10 (invert mount)**

Path-Y (lateral) is parallel to (X) of robot base
 Path-Z (vertical) is parallel to (-Z) of robot base

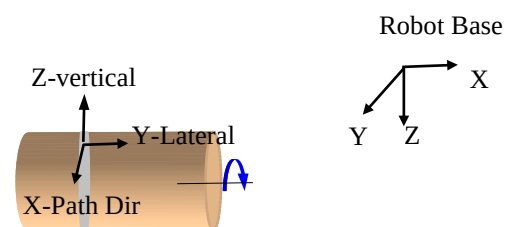
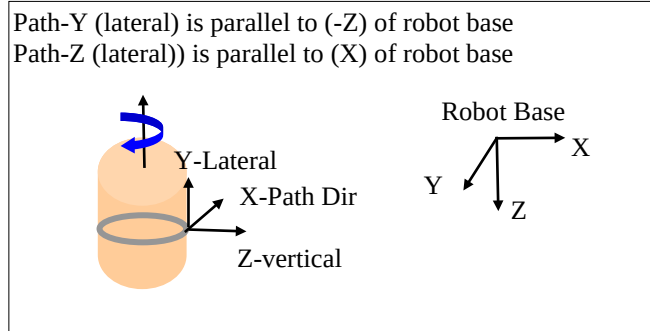
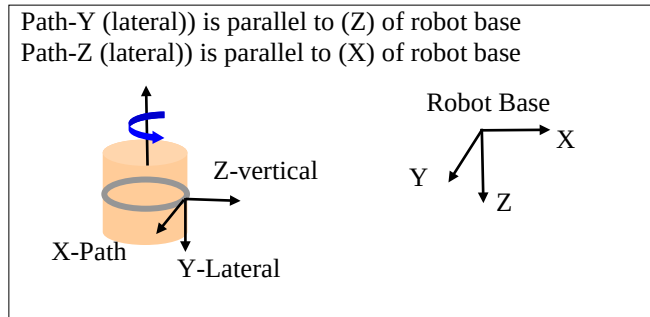


Figure 11 \$cell_cfg = 11 (invert mount)**Figure 12 \$cell_cfg = 12 (invert mount)**

5.3 Auto Setup \$cell_cfg

Starting from v8.30P/16 release, the cell configuration (\$avst_trcst[sch_no].\$cell_cfg) will be setup automatically at the start of the tracking. The auto setup can be disabled by setting the bit0 of \$avst_trctr.\$e2.

6. Tracking Bias Specification

The system variables \$avst_trcst[sch].\$bias_y and \$avst_trcst[sch].\$bias_z allow addition of a constant bias to the track path in lateral or vertical direction respectively. The direction of tracking bias is defined as follows:

- Positive \$bias_y will move TCP toward positive path y.
- Negative \$bias_y will move TCP toward negative path y.
- Positive \$bias_z moves TCP toward weld joint.
- Negative \$bias_z moves TCP away from weld joint.

In UTOOL relative defined cell configurations, the desired vertical compensation is based on sensor scan position in relative to rotation center, only a minor adjustment on vertical bias is needed. However in the UFRAME relative defined cell configurations, the desired vertical compensation is based on the sensor scan position in relative to the robot base. An extra vertical bias is needed to offset the difference between scan position and TCP position due to the sensor lead angle (\$sr_ang) and curvature of the pipe.

6.1 Steps to determine \$bias_z for UFRAME relative cell

Following is the general guideline to determine an initial vertical bias when cell configuration is UFRAME relative (\$avst_trcst[sch_no].\$i2 = 1).

1. Set \$avst_trctp.\$max_ptp.\$z = 0. This will disable the vertical compensation
2. Run stationary tracking program for few tracking segments
3. Check the value of
 - a. \$avst_trctr.\$cur_tcp, TCP position w/r world
 - b. \$avst_trctr.\$pos_jnt, Sensor scanned joint position w/r world
4. Determine the vertical direction based on the cell configuration
5. Set \$bias_z as
 - a. \$avst_trctr.\$pos_jnt (vertical) - \$avst_trctr.\$pos_jnt (vertical) if vertical direction is in positive direction of the major axis
 - b. $-(\$avst_trctr.\$pos_jnt (vertical) - \$avst_trctr.\$pos_jnt (vertical))$ if vertical direction is in negative direction of the major axis
6. Reset \$avst_trctp.\$max_ptp.\$z = 20
7. Run stationary tracking program again (from start)
8. Adjust \$bias_z if needed

6.1.1 Example 1 (cell_cfg = 2)

- \$avst_trctr.\$cur_tcp = < 1104.419, -369.293, 35.971 >
- \$avst_trctr.\$pos_jnt = < 1152.892, -371.079, 21.953 >
- Vertical direction is in World +Z
- \$bias_z = 21.953 – 35.971 = -14.2

6.1.2 Example 2 (cell_cfg = 3)

- \$avst_trctr.\$cur_tcp = < 499.836, 1059.550, 440.962 >
- \$avst_trctr.\$pos_jnt = < 498.763, 1109.245, 428.241 >
- Vertical direction is in World +Z
- \$bias_z = 428.241 – 440.962 = -12.71

6.1.3 Example 3 (cell_cfg = 6)

- $\$avst_trctr.\$cur_tcp = < 981.859, 1065.053, 914.054 >$
- $\$avst_trctr.\$pos_jnt = < 997.340, 1015.748, 918.879 >$
- Vertical direction is in World -X
- $\$bias_z = - (997.340 - 981.859) = -15.581$

7. Stationary Tracking Calibration

7.1 Ten Point Sensor Calibration.

The normal 10 point calibration of the sensor frame is required for stationary track. Refer to the laser sensor frame calibration section of the ArcTool Setup & Operations Manual.

7.2 Sensor Leading Angle Calibration

Stationary tracking requires calibration for sensor leading angle. Follow the steps below you can determine the sensor leading angle in degree:

1. Move robot to a desired starting welding joint location (or a search found location) and mark this location on the pipe as A as shown by figure 7, and record joint angle of the pipe at this location.
2. Turn on laser sensor and mark sensor scan line on the pipe as B as shown by figure 7.
3. Rotate the pipe until mark B aligns to TCP position and record pipe joint angle.
4. Record the difference angle of (angle at B – angle at A).
5. Set this value including the sign to $\$avst_trcst[sch].\sr_ang .

Note the leading angle can be plus or minus. Incorrect sign of leading angle can result in alarm of MIGE-203 Incorrect SR angle and inaccurate value of leading angle can also introduce track error. The sensor leading angle relates to sensor mounting, pipe radius and stationary tracking position. Any operation that involves re-mounting sensor or changing of pipe radius or changing of stationary tracking position would require recalibration of sensor leading angle.

The plots below depict different stationary weld positions at 12 O'clock, at 11 O'clock and at 1 O'clock respectively.

Figure 13 Stationary Position at 12 o'clock

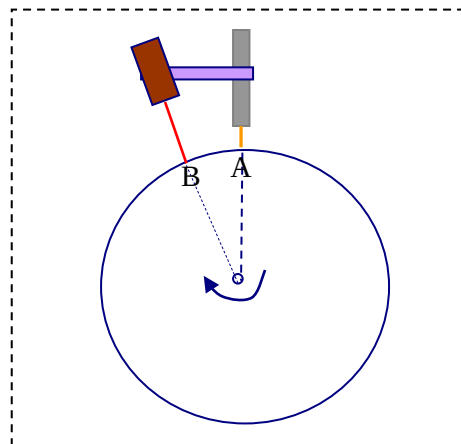
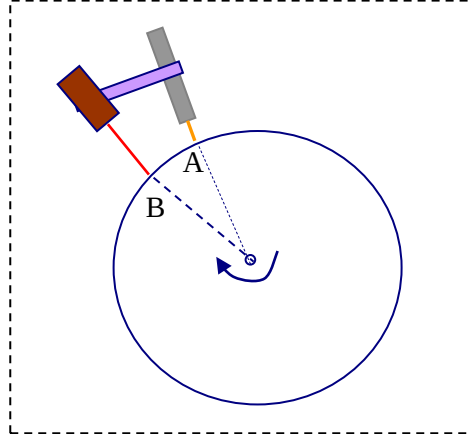
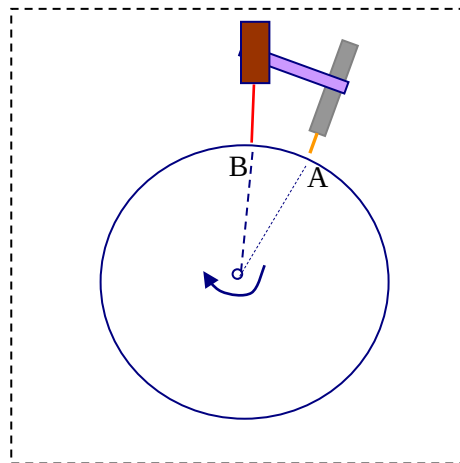


Figure 14 Stationary Position at 11 o'clock**Figure 15 Stationary Position at 1 o'clock**

7.3 Coordinated Motion Setup

Stationary track requires coordination motion option. It can support multiple coordinated pairs. The positioner can be 1-axis or 2-axes positioner.

7.3.1 1-Axis Positioner

The positioner axis for CD pair must be setup as

- The axis type is set to Rotary Axis
- The direction is set to +Z or -Z.

Figure 16 SETUP Coord Sreen (single axis positioner)

```
SETUP Coord
Unknown type calibration Coord Pair:
1
Group Number Leader: 3 Follower: 1
X: 1094.056 Y: 1024.069 Z: 336.571
W: 89.122 P: -3.212 R: -178.526
Axis Number: 1 (Total: 1)
Axis Type: ROTARY
Axis Direction: +Z
Point 1: USED
Point 2: USED
Point 3: USED
```

7.3.2 2-Axes Positioner

Stationary coordinated tracking also supports 2-axes ARC positioner (H871). The axis 2 is the rotational axis. User can set axis 1 to $\pm 90^\circ$ or 0° to form 1G or 2G weld configuration as needed. The default setting of ARC positioner is follows:

Figure 17 2-Axes Positioner SETUP for axis 1

```
SETUP Coord
Unknown type calibration Coord Pair:
1
Group Number Leader: 2 Follower: 1
X: 1125.013 Y: 26.952 Z: 315.726
W: 1.634 P: -0.569 R: 91.532
Axis Number: 1 (Total: 2)
Axis Type: ROTARY
Axis Direction: +X
Point 1: USED
Point 2: USED
Point 3: USED
```

Figure 18 2-Axes Positioner SETUP for axis 2

SETUP Coord	
Unknown type calibration Coord Pair: 1	
Group Number Leader: 2 Follower: 1	
X: 1125.013 Y: 26.952 Z: 315.726	
W: 1.634 P: -0.569 R: 91.532	
Axis Number: 2 (Total: 2)	
Axis Type:	ROTARY
Axis Direction:	Z
Point 1:	USED
Point 2:	USED
Point 3:	USED

For other general 2-axes positioners, the axis direction for a rotational axis, normally axis 2, should be set to 5 (Z) or 6 (-Z). The angle for axis 1 should be set properly such that the pipe rotation axis is parallel to one of the major robot axis in $\pm X$, $\pm Y$, or $\pm Z$.

7.4 Sensor Scan and Registration Rate Control

The sensor scan rate is determined by `$sr_setup.$update_time`, normally, it is set to 50ms or 100ms. The scan data will be stored in the stationary track buffer. For slow travel speed it would require a big buffer to hold the scan data for sensor leading angle. Practically, it is unnecessary to have too many scan data over a small pipe rotation angle range.

In CD stationary track, the tracking buffer size is fixed to 256. By default, the scan data is stored to track buffer only if an incremental pipe rotation angle exceeds a threshold. The default threshold is determined by $|\$sr_ang|/tracking_buffer_size$. For example, for 25 degree of sensor leading angle and 256 of buffer size, the threshold would be about 0.1 degree.

However, if you would like to register each scan data into the tracking buffer you can set `$avst_trctr.$scan_mode` to 1. In this case each scan data will be stored in the tracking buffer.

8. Tack Avoidance and Control

When laser scanner fails to find a valid joint due to a tack or other factors such as noise or reflection of joint surface, the coordinated stationary tracking will continue using the last valid offset until a valid joint is found. User can enable CD stationary tracking tack function by setting `$stack_enb = TRUE` to obtain additional functions for the tack recognition, measurement and tack control.

8.1 Tack Recognition

The system variable `$stack_min_l` (in degree) defines a minimum tack threshold. Only if a consecutive number of failed scans over a tack section exceed this threshold, the bad scan section is considered as a valid tack. There is an exception when failure scan occurs on a welded joint; the consecutive scan failures are anticipated. To avoid the MIGE-206 alarm at the end of the pipe weld, user must set start weld joint angle (including sign) to `$f1`.

8.2 Tack Length Measurement

Once a valid tack is detected the system will output the measured tack length to \$stack_length (in degree). This value represents most recent tack length in degree.

8.3 Tack Alarm

When a valid tack length exceeds the value of \$stack_max_l (in degree) or sensor lead angle (\$sr_ang), system will post an abort alarm MIGE-206 “Tack exceeds limit” to stop tracking operation.

8.4 Real Time Tack Control

The tack avoidance control also monitors if current TCP is over the tack section or not in real time. The value of \$stack_pres is set when tack is present to current TCP. User can use this value for real time tack control.

Figure 19 indicates a valid tack is detected but tack has not been reached to TCP yet.

Figure 19 TACK is detected

$\$min_tack_l < \$stack_length < \$max_lack_l$.
Valid tack but TCP is not at tack section,
\$stack_pres = FALSE

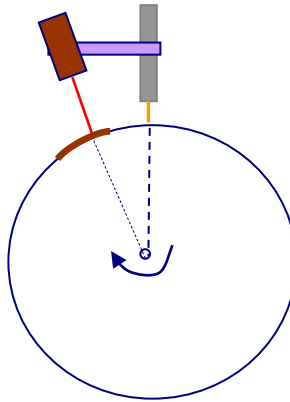
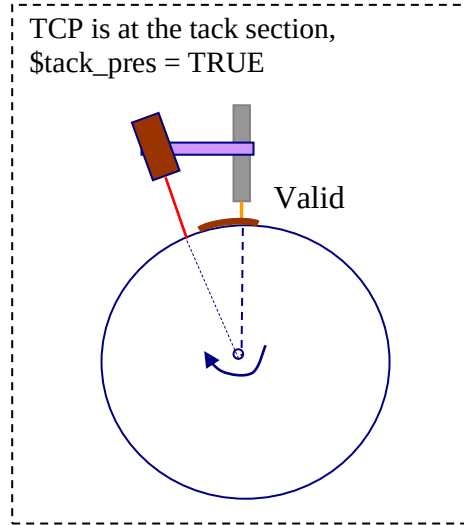


Figure 20 indicates TCP is on the tack section.

Figure 20 TCP at TACK Section



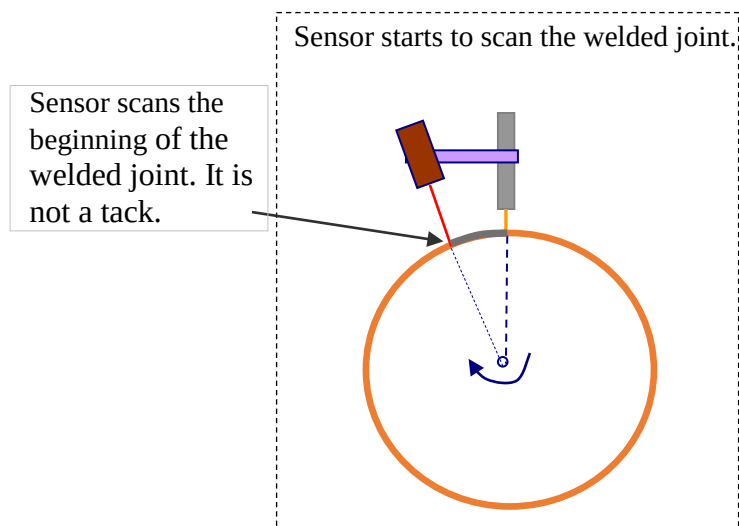
8.5 Overlap Welding Control

At the overlap joint welding, the system would take two actions:

1. Does not post MIGE-026 alarm because the sensor scans the welded joint.
2. The tracking is automatically turned off but the last offset is applied to the overlap welding section.

Figure 21 indicates that sensor starts scanning welded joint when pipe rotation angle closes to a full turn. For the remaining of the welding process, the sensor can't detect the joint continuously, but the system will not post MIGE-026 alarm.

Figure 21 Scans at welded joint



9. Coordinated Stationary Tracking Setup

Coordinated stationary tracking function supports up to 10 tracking schedules. You can setup stationary tracking by system variable \$AVST_TRCTR and \$AVST_TRCST[sch]. The \$AVST_TRCTR is for general setup and the \$AVST_TRCST[sch] is for schedule setup. The array index for \$avst_trcst[sch_no] corresponds to a sensor schedule. The tables below provide detail descriptions for each item. The yellow marked items are setup related items, the green are output for monitoring purpose.

Table 1 \$avst_trcst[sch_no]

\$cell_cfg Default 1	This item specifies cell configurations: UTOOL relative supports following 6 configurations: 1. Path y is parallel to world -y 2. Path y is parallel to world y 3. Path y is parallel to world -x 4. Path y is parallel to world x 5. Path y is parallel to world -z 6. Path y is parallel to world z UFRME relative supports following 12 configurations: 1. Path y, z are parallel to world -y, z 2. Path y, z are parallel to world y, z 3. Path y, z are parallel to world -x, z 4. Path y, z are parallel to world x, z 5. Path y, z are parallel to world -z, -x 6. Path y, z are parallel to world z, -x 7. Path y, z are parallel to world y, -z 8. Path y, z are parallel to world -y, -z 9. Path y, z are parallel to world x, -z 10. Path y, z are parallel to world -x, -z 11. Path y, z are parallel to world z, x 12. Path y, z are parallel to world -z, x
\$ldr_gnum Default 2	Leader group number, base 1.
\$ldr_axis Default 1	Leader rotation axis, base 1.
\$sr_ang Default 25	Sensor lead angle in degree. See the calibration for how to determine this value. This value can be plus or minus.
\$bias_y Default 0	Bias y in mm relative to weld path. A positive value moves TCP toward positive path y and a negative value will move TCP toward negative path y.
\$bias_z Default 0	Bias z in mm relative to weld path. A positive value will move TCP toward to weld joint and a negative value will move TCP away from weld joint.
\$avebef Default 4	Average scan number before current sensor data.
\$aveaft Default 4	Average scan number after current sensor data.
\$stack_enb	Enable pipe tack control.
\$stack_pres	True when TCP at tack section, FALSE when TCP at regular joint.

Table 1 \$avst_trcst[sch_no]

\$stack_min_l	Minimum tack length in deg. When tack length is less than this value, this section will not be marked as tack.
\$stack_max_l	Maximum tack length in deg. When tack length exceeds this value, system will post TACK Exceeds Limit alarm if \$stack_enb is TRUE.
\$stack_length	System output real time measured tack length in degree.
\$i1	This item specifies two different stationary control methods 0: default, non-segment based control, work with weave operation 1: segment based control, work w/o weave operation.
\$i2	This item expands the cell configurations 0: TOOL relative (default) supports cell_cfg of 1 – 6. 1: UFRAME relative supports cell_cfg of 1 – 12.
\$f1	Pipe welding start angles (leader group) including the sign.
\$f2	Servo delay compensation for segment based high speed tracking. Value range (0 – 500) in ms. The default value is 0.

Table 2 \$avst_trctr

\$ena	TRUE: enable stationary tracking function. FALSE: disable stationary tracking function. The change is effective without re-power of the controller.
\$sch	Current track schedule.
\$gap	Current gap in mm.
\$max_inc Default (0.0, 0.05, 0.05)	Max allowed ITP offset, path relative. Positive number. Set to zero will disable tracking.
\$max_pth Default (0.0, 10.0, 20.0)	Max allowed path offset, path relative. Positive number. Set to zero will disable tracking.
\$cur_tcp	Current TCP position.
\$pos_jnt	Current scan joint location w/o bias in robot base.
\$pos_cd	Pipe rotation center in world frame.
\$pos_wd	Stationary position in world frame.
\$ofs_pth	Actual total applied offset in path relative.
\$ofs_wld	Total track offset in world.
\$ofs_rpm	RPM offset at pitch.
\$safe_dt Default: 0.4	Max allowed change for rpm in vertical direction at new pitch point. It will not clamp for lateral rpm offset.
\$overlap	Internal use.
\$errbit	Internal use.
\$scan_mode Default 0	0: default, register sensor scan data when pipe rotation angle > \$sr_ang/256. 1: register each sensor scan data.
\$lsq_dat_no Default 10	Least square smooth sample no. Valid range is [5, 10]. Set this value < 5 will disable Least Square Filter.
\$lsq_ord_no Default 2	Least square smooth order no.

10. RPM/MP Clock Function Specifications and Restrictions

- RPM/MP clocking function is effective only if the Coordinated Stationary Tracking function is enabled (\$avst_trctr.\$ena = TRUE).
- RPM/MP clocking is motion segment based.
- RPM/MP clocking is limited to maximum 80 rpm motion segments.
- Each motion segment has same rotation angle for the leader group except for the overlap segment.
- Support clockwise or counter-clockwise clocking.
- RPM/MP clocking is permitted ONLY for a fix stationary position.
- Support overlap for multiple pass. The offset of the last clocked segment is applied to the overlap segment.
- RPM supports distance or time based pitch mode
 - At distance pitch mode (\$rpm_pg.\$pitch_mode = 0), the \$pitch is in degree.
 - At time pitch mode (rpm_pg.\$pitch_mode = 1), the \$pitch is in millisecond.

In order to use the RPM Clock function, the tracking weld pass must use the same torch position as the multiple pass. For example, if the robot pass weld is accomplished with the weld torch at the 10:00 O'clock position, the multi-pass must also be welded with the torch at 10:00 O'clock position. It is common in open root pipe welding for the first (root) pass to be welded with torch at the 10:30 position in order to optimize the effect of gravity on the weld pool. Subsequent passes are typically welded with the torch located near the 12:00 O'clock position. In order to clock the RPM offsets, it is necessary to track on the second weld pass and use RPM Clocking on subsequent passes.

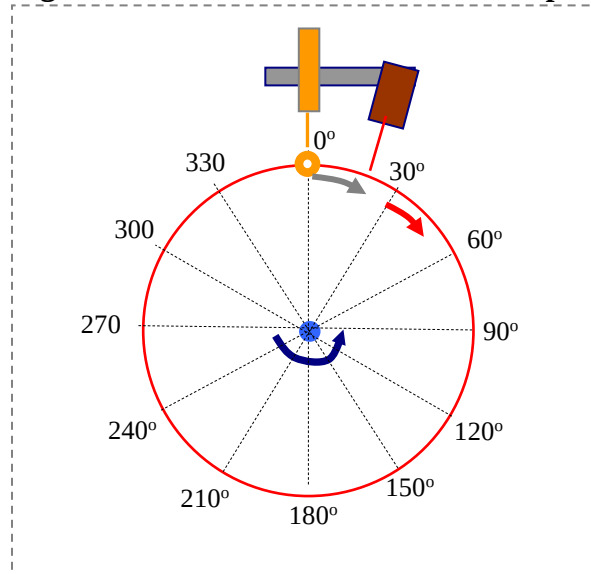
11. RPM/MP Clock Operation

Following examples illustrate the usage of the clock function for multiple pass welding. In these examples, leader segment has 5° rotation angle.

11.1 Example 1 torch at 12 clock position

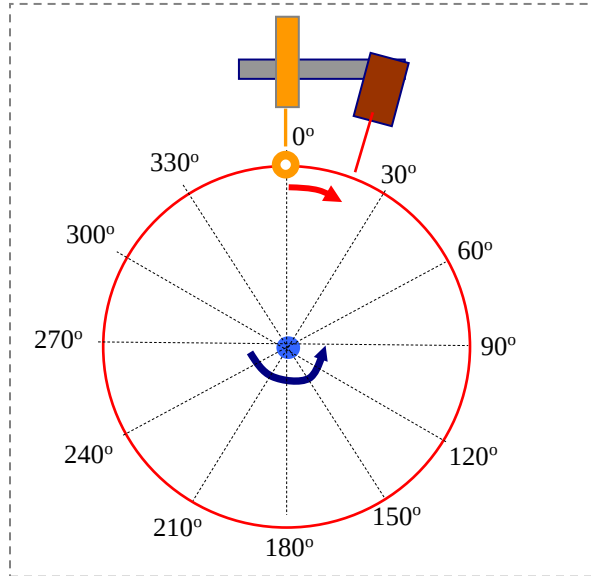
- Root pass: stationary track position at 0°, tracking starts at 0° and **weld starts at 30°** due to sensor leading angle. The welding ends at 392° with 2° of overlap as shown by figure below.

Figure 22 Root Pass, Torch at 12 clock position



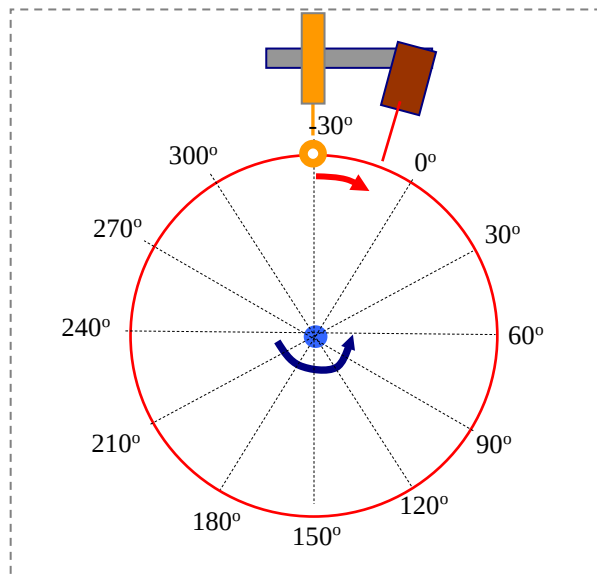
- Multi-pass-1: The rpm clock = 0 and weld starts at 0° and ends at 362° with 2° of overlap.

Figure 23 MPass 1, Torch at 12 clock position



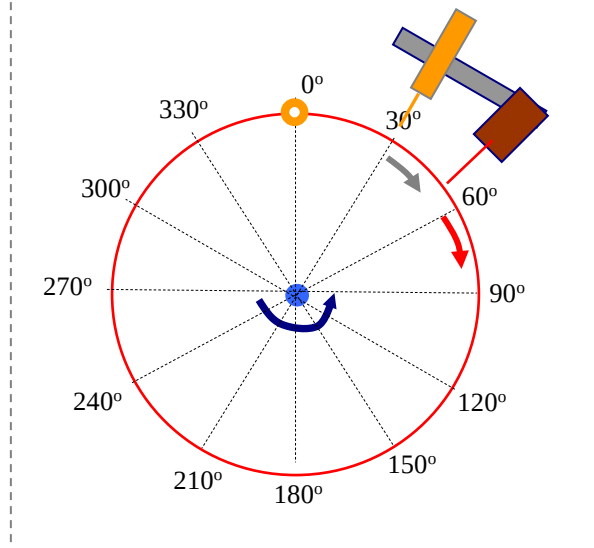
- Multi-pass-2: The rpm clock = -6 (equivalent to -30° rotation), weld starts at -30° and ends at 332° with 2° of overlap.

Figure 24 MPass 2, Torch at 12 clock position

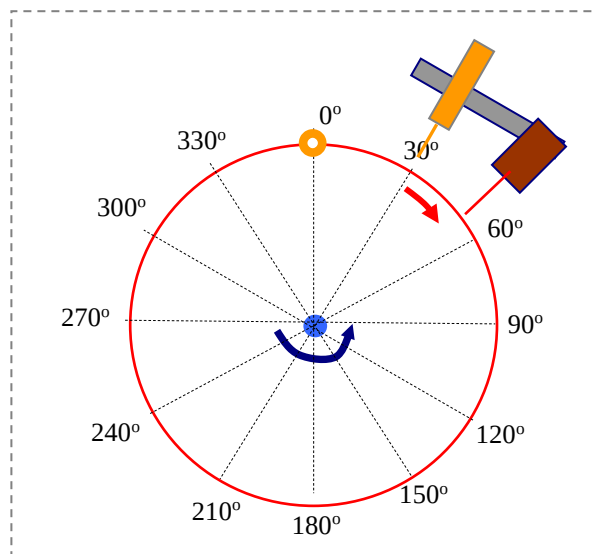


11.2 Example 2 torch at 1 clock position

- Root pass: stationary track position at 30° , tracking starts at 30° and **weld starts at 60°** due to sensor leading angle. The welding ends at 422° with 2° of overlap as shown by figure below.

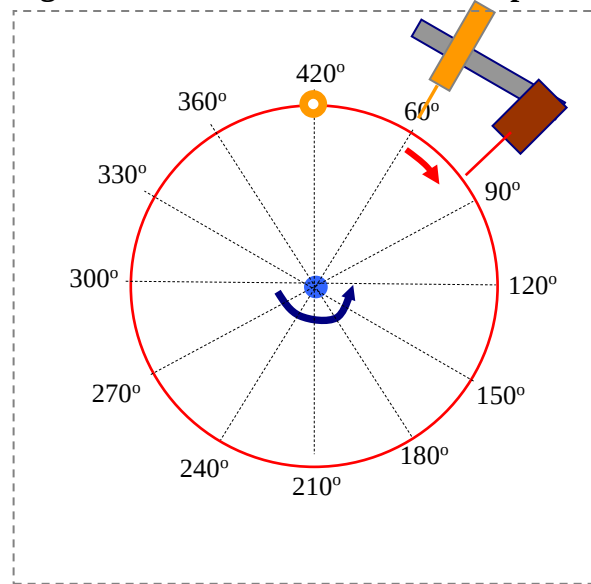
Figure 25 Root Pass, Torch at 1 clock position

- Multi-pass-1: The rpm clock = 0, **weld starts at 30°** and ends at 392° with 2° of overlap.

Figure 26 MPass 1, Torch at 1 clock position

- Multi-pass-2: The rpm clock = 6 and weld starts at 60° and ends at 422° with 2° of overlap.

Figure 27 MPass 2, Torch at 1 clock position



12. RPM/MP Clock Setup

RPM/MP clocking function supports up to 10 RPM buffers. You can setup the RPM/MP clocking function by setting system variable \$rpm_clk[buf_no]. The array index corresponds to RPM buffer number. The yellow mark items are input, green mark items are output.

Table 3 \$rpm_clk[buf_no]

\$enable	Enable/disable clocking function for this rpm buffer. Must be enabled at root track pass if clocking is desired for multiple passes.
\$clk_sg Default 0	RPM segment clocking index. The value can be plus or minus. Default 0 means no RPM/MP clocking.
\$lhd_sg_t Default 6	Number of look ahead segments used at root pass.
\$olp_sg Default 1	0: no overlap segment, 1: one overlap segment.
\$seg_ang	Leader rotation angle per segment for internal use.
\$tot_sg	Total rpm segments for internal use.
\$sghp	Internal data.

13. Coordinated Stationary Tracking and RPM/MP Clock Programming

In stationary tracking, the robot position is a single point and pipe rotation angle is incremental during coordinated stationary tracking. The entire program contains few sections:

- A search section to find initial start position as a stationary tracking point for the robot.
- Generate all programming points stored in position register automatically by calling a utility routine PR_CLK_G13.TP. This routine contains 7 arguments:
 1. Start Position Register Number
 2. Start pipe angle
 3. End pipe angle
 4. Incremental angles
 5. RPM/MP clocking index.
 6. Search Position Register Number
 7. Overlap angles
- Tracking only section. This section is tracking for \$SR_ANG angle to pre-fill the tracking buffer data without welding.
- Tracking and welding section. Arc Start after sensor pre-filled the sensor data into the tracking buffer.
- Arc End when pipe rotates a full turn plus \$sr_ang and possible overlap angles.
- The multi-pass section.

13.1 PR_CLK_G13 Utility TP

This utility routine is used to generate all program points for stationary tracking and multi-pass TP program. For tracking program the value for RPM/MP clocking index must be 0. In this example, the leader group is in group 3 with 1-axis positioner. The user can create their own utility routine based on actual leader group number and rotation axis number for the positioner.

Table 4 PR_CLK_G13 Utility TP

<pre> /MN 1: !AR[1] str PR# ; 2: !AR[2] str ang ; 3: !AR[3] end ang ; 4: !AR[4] int ang ; 5: !AR[5] clk_idx ; 6: !AR[6] src PR# ; 7: !AR[7] ovl ang ; 8: R[101]=AR[1] ; 9: R[102]=AR[2] ; 10: R[103]=AR[3] ; 11: R[104]=AR[4] ; 12: R[105]=AR[5] ; 13: R[106]=AR[6] ; 14: R[107]=AR[7] ; 15: !total ang ; 16: R[110]=R[103]-R[102] ; 17: !total idx ; 18: R[110]=R[110]/R[104] ; 19: R[113]=R[104]*R[105] ; 20: R[102]=R[102]+R[113] ; 21: !set str ang ; 22: PR[R[101]]=PR[R[106]] ; 23: !set str ang ; 24: PR[GP3:R[101],1]=R[102] ; 25: FOR R[111]=1 TO 80 ; 26: IF R[111]>R[110],JMP LBL[10] ; 27: R[112]=R[111]+R[101] ; 28: PR[R[112]]=PR[R[101]] ; 29: R[113]=R[111]*R[104] ; 30: PR[GP3:R[112],1]=R[102]+R[113] ; 31: ENDFOR ; 32: LBL[10] ; 33: IF R[107]=0,JMP LBL[11] ; 34: R[112]=R[112]+1 ; 35: PR[R[112]]=PR[R[101]] ; 36: PR[GP3:R[112],1]=R[102]+R[113]+R[107] ; 37: LBL[11] ; /POS /END </pre>	- Note the use of the Group mask - Note the use of the Group mask - Note the use of the Group mask
--	---

13.2 Root Pass Program Example

Following program template is used for

- 5 degree segment length for rotational axis in the leader group
- 6 look ahead tracking ONLY segments
- 72 welding segments
- 1 overlap segment
- Total 79 coordinated tracking segments

Table 5 Search & Tracking for Root Pass Section

<pre> /MN 1:J P[1] 20% FINE ; 2: SENSOR ON[1]; 3:J P[2] 10% FINE ; 4: SENSOR SEARCH START PR[198]; 5: SENSOR SEARCH POINT[10] ; 6: SENSOR SEARCH END ; 7: PR[201]=PR[198] ; 8:L PR[201] 100mm/sec FINE ; 9: CALL PR_CLK_G13(201,0,390,5,0,201,2); 10: PAUSE ; 11: Sample Start[1]; 12: LOCK PREG ; 13: Track SENSOR[10] RPM[8]; 14:L PR[202] WELD_SPEED CNT100 COORD ; 15:L PR[203] WELD_SPEED CNT100 COORD ; 16:L PR[204] WELD_SPEED CNT100 COORD ; 17:L PR[205] WELD_SPEED CNT100 COORD ; 18:L PR[206] WELD_SPEED CNT100 COORD ; 19:L PR[207] WELD_SPEED CNT100 COORD : Arc Start[1] ; 20: Weave Sine[2]; 21:L PR[208] WELD_SPEED FINE COORD ; 22:L PR[209] WELD_SPEED CNT100 COORD ; 23:L PR[210] WELD_SPEED CNT100 COORD ; 24:L PR[211] WELD_SPEED CNT100 COORD ; ... 89:L PR[276] WELD_SPEED CNT100 COORD ; 90:L PR[277] WELD_SPEED CNT100 COORD ; 91:L PR[278] WELD_SPEED CNT100 COORD ; 92:L PR[279] WELD_SPEED CNT100 COORD ; 93:L PR[280] WELD_SPEED FINE COORD : Arc End[1] ; 94: Track End ; 95: Weave End ; 96: Sample End ; 97: UNLOCK PREG ; 98: WAIT 1.00(sec); 99:J P[1] 20% FINE ; /END </pre>	<div data-bbox="993 703 1222 829" data-kind="parent" data-rs="2"> } Search </div> <div data-bbox="993 850 1524 913" data-kind="parent" data-rs="2"> } Call to create position register data </div> <div data-bbox="993 1018 1349 1144" data-kind="parent" data-rs="4"> } 6 segments (0-30) for track only section </div> <div data-bbox="993 1270 1354 1428" data-kind="parent" data-rs="4"> } 72 segments (30-390) for track and weld section </div> <div data-bbox="993 1501 1390 1596" data-kind="parent" data-rs="2"> } last segment (390-392) for overlap weld section </div>
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13.3 Multiple Pass Program Example

Prior to execution of multi-passes, you must set the value for \$lhd_sg_t and \$clk_sg in \$rpm_clk. The value of \$lhd_sg_t corresponds to the number of look ahead segments used in the root tracking pass and the value of \$clk_sg specifies the number of pipe rotation segments for the starting welding position.

Table 6 Multiple Pass Welding Section

```

/MN
1:J P[1] 20% FINE ;
2:J P[2] 10% FINE ;
3: PR[201]=PR[198] ;
4: $RPM_CLK[8].$CLK_SG=5 ;
5: R[2]=$RPM_CLK[8].$CLK_SG ;
6: CALL PR_CLK_G13(201,0,360,5,R[2],201,2) ;
7: LOCK PREG ;
8: MP Offset PR[11] RPM[8] ;
9:L PR[201] WELD_SPEED FINE COORD
: Arc Start[1] ;
10: Weave Sine[3] ;
11:L PR[202] WELD_SPEED CNT100 COORD ;
12:L PR[203] WELD_SPEED CNT100 COORD ;
13:L PR[204] WELD_SPEED CNT100 COORD ;
14:L PR[205] WELD_SPEED CNT100 COORD ;
15:L PR[206] WELD_SPEED CNT100 COORD ;
19:L PR[207] WELD_SPEED CNT100 COORD ;
...
78:L PR[269] WELD_SPEED CNT100 COORD ;
79:L PR[270] WELD_SPEED CNT100 COORD ;
80:L PR[271] WELD_SPEED CNT100 COORD ;
81:L PR[272] WELD_SPEED CNT100 COORD ;
82:L PR[273] WELD_SPEED CNT100 COORD ;
83:L PR[274] WELD_SPEED FINE COORD
: Arc End[1] ;
84: MP Offset End ;
85: UNLOCK PREG ;
86: Weave End ;
87: WAIT 1.00(sec) ;
88:J P[1] 20% FINE ;

```

14. Alarms

14.1 Alarms for CD Stationary Tracking

Table 7 Stationary Tracking Alarms

Alarm	Severity	Description
MIGE-200 NO COORD Motion error	Abort	Stationary track is enabled for non-coordinated tracking motion.
MIGE-201 Can't find tracking data	Abort	Check cell configuration and calibration. If this occurs at middle of the tracking and current value of \$avst_trctr.\$scan_mode is 1, you should change \$scan_mode to 0.
MIGE-202 Incorrect leader group	Abort	Schedule \$ldr_gnum does not match actual COORD leader group.
MIGE-203 Incorrect SR angle	Abort	Either the value or the sign is incorrect.
MIGE-204 Max-L Offset limit	Warning	Reached max lateral offset limit.
MIGE-205 Max-V Offset limit	Warning	Reached max vertical offset limit.
MIEG-206 Tack exceeds limit	Abort	Tack length (deg) exceeds the limit.

14.2 Alarms for RPM/MP Clocking

Table 8 PRM/MP Clocking Alarms

Alarm	Severity	Description
RPM-050 Max segment limit	Abort	This error is posted if clock function is enabled and the recorded segment number reaches its limit. The limit is 80 for clock function.
RPM-051 Stationary TRK disabled	Abort	This error is posted if clock function is enabled but the stationary tracking is disabled in root pass. Clocking function requires enable of stationary track function.

15. Setup Existing Sensor Schedule & Search Schedule

The existing Sensor Track Schedule values are not used when stationary tracking is enabled except for the SR Track Joint Num and Tracking Type. The Tracking Type should be set to Standard for the coordinated stationary tracking.

Note: don't set Tracking Type to Stationary. The existing Sensor Search Schedule is always active when searching is done. For reference, the existing Sensor Schedule detail menu:

1 Sensor Schedule: 1 []

Tracking Setup

2 SR Track Joint Num	1
3 Tracking Type	Standard
4 Motion Sensitivity	3
5 Y-Bias (mm)	0.00
6 Z-Bias (mm)	0.00
7 Stationary Track Frame	0
8 Tack Avoidance	Disable
9 Tack Threshold(mm)	2.00
10 Tack Length (mm)	40.00
11 Joint End Detect	Disable
12 Start Distance (mm)	100.00
13 Detect Do Index	0
14 Joint End OFS (mm)	0.00

Search Setup

15 SR Search Joint Num	1
16 Search Type	Position
17 Detection Count	1
18 Retry Count	0
19 Retry Direction	X
20 Retry Distance (mm)	3.0
21 Y-Bias (mm)	0.00
22 Z-Bias (mm)	0.00
23 Search Frame	UFRAME
24 Dynamic Search	Disable
25 Search Dist (mm)	100.00
26 Search Spd (mm/sec)	100.00

Green highlight indicates active parameters. All others not used when Stationary Tracking is enabled.